

# MOHAMED HANY REDA

AI / Machine Learning Engineer · Computer Vision · Reinforcement Learning

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## PROFILE

Machine-learning engineer with 3+ years building AI products and shipping them as real applications, not notebooks — computer vision, reinforcement learning and explainable AI. I take a model from data through training and evaluation to a running product, and I bring full-stack engineering (Python, FastAPI, Docker) so the model actually reaches users. Available remotely worldwide, on-site or hybrid; open to relocation.

## SELECTED HIGHLIGHTS

- Shipped a published **medical-imaging AI**: EfficientNet-B0, 90.21% accuracy, Grad-CAM explanations, delivered as a Streamlit app and Telegram bot with automated PDF reports.
- Built a 400-lane logistics **digital twin with PPO routing and multi-agent CTDE** reinforcement learning, modelling ~96,400 items/hour, with 530+ tests.
- Trained **safe reinforcement-learning drone control** (PPO/LSTM) under domain randomisation and fault injection; exported to ONNX.
- Built an **evolutionary racing lab** comparing PPO/SAC/A2C across procedurally generated worlds, with saliency and activation explainability.
- Delivered the full **deployment layer** around each model — apps, APIs, dashboards — so the AI ships.

## SKILLS

**AI / ML:** PyTorch, computer vision (EfficientNet, OpenCV), reinforcement learning (PPO/SAC/A2C, multi-agent CTDE), explainable AI (Grad-CAM, saliency), LLM tooling

**ML engineering:** Python, training/eval loops, data pipelines, ONNX, experiment logging, reproducible runs, Pytest

**Deployment:** FastAPI, Streamlit, Docker, SQLite/PostgreSQL, Telegram bots, PySide6 dashboards

**Also full-stack:** JavaScript, React, Flutter/Dart, Node.js/Firebase — I ship the product around the model

**Delivery:** Requirements, agile, team leadership up to 10, budget ownership

## EXPERIENCE

### Full-Stack Developer / Technical Owner | HMS

January 2024 — Present

- Own the corporate web product end to end — requirements from stakeholders, architecture, front and back end, release and support.

### Full-Stack Developer / Product Delivery Owner | AuraGift

January 2024 — Present

- Own the client side and UX of a live premium e-commerce platform; repeatable Firebase deployments with no downtime.

### Technical Project Manager / Product Engineer | Freelance

June 2022 — Present

- Single point of accountability from discovery and SRS through architecture, build and handover.
- Led a team of 10 on an AI logistics system, ~\$13,000 budget, 11 months.

### Technical Project Lead / Full-Stack Developer | HMS

January 2023 — June 2023

- Led a team of 8 and managed a ~\$10,000 budget while contributing to the build.

### Unreal Engine Developer / Independent Product Owner | Indie

January 2020 — Present

- Networked multiplayer with server authority and cloud player data; 3 published Android titles.

### Media Team Lead | Resala Charity Organization, Egypt

January 2020 — December 2020

- Led the media function; completed a six-month leadership and management programme.

## KEY PROJECTS

- Fracture Detection AI** — EfficientNet-B0 + Grad-CAM on 4,083 X-rays; 90.21% accuracy; Streamlit + Telegram. Research software, not a certified medical device.
- Sorting-Center Digital Twin** — PPO routing, multi-agent CTDE, discrete-event simulation; ~96,400 items/hour, 530+ tests.
- Strict Drone Safe-RL** — PPO/LSTM under domain randomisation and fault injection; ONNX export. Simulation/research platform only.
- AI Evolution Racing Lab** — PPO/SAC/A2C tournaments across procedural worlds, with saliency/activation explainability.
- Multi-Agent Air-Defense Simulation** — cooperating agents, self-play, trajectory prediction, PySide6 live dashboard.

## EDUCATION

**Bachelor — Computer Science & Engineering**, Tomsk State University of Control Systems and Radioelectronics (TUSUR) · 2026 · Unreal Engine course author (Udemy, 2024) · Digital Marketing (Google, 2021)

## LANGUAGES

**Arabic** — Native · **English** — C2, full professional · **Russian** — B2, upper-intermediate